# MILITARY SPECIFICATION

# AMPLIFIERS, AUDIO FREQUENCY;

# AND AMPLIFIER-CONTROL GROUPS—SHIPBOARD ANNOUNCING

## 1. SCOPE

1.1 This specification covers audio frequency amplifiers and controls, shipboard announcing type, for use aboard Naval ships. The above amplifiers are to be employed as units in shipboard announcing systems which disseminate alarm signals, commands, and information as required in the tactical operation of Naval surface ships and submarines.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-T-27 | - Transformers and Inductors (Audio, Power, and High Power Pulse) General Specification for. |
| MIL-I-983 | - Interior Communication Equipment, Naval Shipboard; Basic Design Requirements for. |
| MIL-E-17362 | - Electronic Repair Parts Requirements, Procedures for Provisioning Technical Documentation and Stock Numbering. |
| MIL-E-17555 | - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-275 | - Printed Wiring for Electronic Equipment. |
| MIL-STD-429 | - Printed Circuit Terms and Definitions. |

DRAWINGS

BUREAU OF SHIPS

| | |
|---|---|
| 9000-S6202-73075 | - Switch-Multipole Rotary, Panel Mounted, Type S-1JR. |
| 9000-S6202-73295 | - Switch-Multipole Rotary, Action Cut-out, Type 4JR. |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Preproduction sample.- Prior to beginning production a sample shall be tested as specified in 4.2 (see 6.2).

3.2 Description.-

3.2.1 Equipment complement.- The equipment shall consist of amplifier assemblies, or amplifier-oscillator assemblies including control facilities, such as switches and relays, mounted in racks or enclosures.



Source: https://assist.dla.mil -- Downloaded: 20261O-19T09:55Z
Check the source to verify that this is the current version before use.

3.3 General features.- The equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.

General Information
Definitions
Materials, general
Substitution of (equal or superior) materials
Fungus-inert materials
Flammable materials
Arc-resistant materials
Toxic materials
Wood
Aluminum
Magnesium
Zinc
Other metals
Springs
Plastics
Ceramics
Impregnating, embedding and encapsulating compounds
Glass
Lubricants and lubrication
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Gaskets
Dials and pointers
Dial sizes
Locking devices
Washers
Parts-electrical-general
Electron tubes
Electron tube or capacitor sockets
Capacitors
Variable resistors
Transformers and inductors
Relays
Electrical tapes
Switches
Indicator lights and lampholders
Fuses
Fuseholders and fuseclips
Enclosures-general
Enclosure-accessibility
Enclosure-degree of
Enclosure mounting
Stiffening grooves
Minimum sheet metal thickness
Through bolting
Ventilation
Size
Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Welding, structural
Temperature and humidity (special cold limit applies)



Source: https://assist.dla.mil -- Downloaded: 20261O-19T09:55Z
Check the source to verify that this is the current version before use.

Shock, vibration and inclination (vital vibration test required)
Primary power supply circuits
Power supply tolerances (normal power supply 115 volts 60 cycles single phase a.c.)
Safety (personnel hazard)
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection
Terminal boards and terminals
Wiring
Color coding
Electrical insulation
Dielectric strength and insulation resistance clearances
Drawings-general
Drawings-preliminary
Drawings-working
Schematic diagrams
Wiring diagrams
Drawing list
Assembly drawings
Bill of materials
Interchangeability and standardization
Manuals
Repair parts (see 3.3.1)
Repair parts (electronic)
Designation and marking
Reports
Workmanship

3.3.1 Repair parts.- Onboard repair parts (equipment repair parts) and stock repair parts, if required by the contract or order, shall be determined, identified and procured in accordance with MIL-E-17362.

3.4 Mechanical requirements.-

3.4.1 Amplifier assemblies and amplifier-oscillator assemblies.-

3.4.1.1 Definition.- An amplifier assembly or amplifier-oscillator assembly shall consist of a basic amplifier unit or an amplifier-oscillator unit complete with chassis and all parts, intended for mounting within an amplifier rack or enclosure, or within a control rack.

3.4.1.2 Construction.-

3.4.1.2.1 Accessibility.- The construction shall be such that all wiring terminals shall be accessible for test purposes without requiring removal of an amplifier assembly or amplifier-oscillator assembly from the rack or enclosure in which it is mounted. (The use of hinged or pivoted chassis secured by readily accessible thumb screws will be acceptable for this purpose.)

3.4.2 Amplifier and control racks.-

3.4.2.1 Definition.- An amplifier or control rack shall consist of an assembly of one or more amplifier or amplifier-oscillator assemblies or a combination of amplifier and amplifier-oscillator assemblies, together with control facilities, such as switches and relays, mounted in a single rack or enclosure.

3.4.2.2 General construction.- The amplifier rack or enclosure or control rack shall be designed for deck or bulkhead mounting. Any rack or enclosure exceeding 250 pounds in weight or 30 inches in height shall be designed for deck mounting. All other racks or enclosures shall be designed for bulkhead mounting.



Source: https://assist.dla.mil -- Downloaded: 202610-19T09:55Z
Check the source to verify that this is the current version before use.

3.4.2.3 Front servicing.- All racks or enclosures shall be designed to permit complete accessibility to all parts and subassemblies from the front of the enclosure.

3.4.2.4 Removable covers and doors.- Racks or enclosures shall be provided with removable covers or doors which may be readily opened and removed for servicing or inspection. Covers or doors shall be secured by captive "quick opening" type fasteners having slotted heads, or by captive slotted knurled head machine screws, capable of being readily unfastened or unscrewed by use of a screwdriver or coin.

3.4.2.5 Location of heavy parts in assemblies.- Heavy parts such as power transformers, output transformers and similar parts shall be mounted in assemblies in such locations that the center of gravity of the assembly will be at, or in close proximity to, its physical center.

3.4.2.6 Mounting of amplifier and amplifier-oscillator assemblies.- Dowels or studs designed to match with bushings or holes in the assembly, or shelves or supports shall be provided to sustain the weight of the assembly while the assembly is being mounted in or removed from the rack or enclosure. Shelves or angle supports shall be used where the weight of the assembly exceeds 50 pounds.

3.4.2.7 Mounting of controls.- All switches, operating controls, test controls and meters shall be mounted on the front of the control rack or panel and shall be visible and accessible without requiring opening of doors or removal of covers.

3.4.2.8 Terminal boards.- Terminal boards shall be provided in a convenient location, not less than 12 inches from the bottom of deck mounted racks, for connection of all ship's cables entering the enclosure. Amplifier racks or enclosures designed for use with a separate control rack or other amplifier racks shall have provisions for inter-rack connections. Where the control rack is separate from amplifier racks all external ship's cables shall be connected in the control rack. Where departure from the above requirements are necessary, specific approval by the Naval Ship Engineering Center (NAVSEC) will be required.

3.4.2.9 Provision for cable entrance.- Removable plates, suitable for drilling to accommodate terminal tubes for cable entrance, shall be provided at the top of the rack or enclosure. Plates for top cable entrance shall be provided with gaskets to prevent water, falling from above, from entering the enclosure. Removable plates suitable for drilling, to receive cable clamps shall be installed at the bottom of enclosures for cables entering from the bottom. These plates need not be provided with gaskets.

3.4.2.10 Weight limit.- The weight of any single component assembly shall not exceed 125 pounds. In the design of large power amplifier assemblies it is permissible to separate the power supply from the amplifier to meet the above requirements.

3.4.2.11 Degree of enclosure.- The degree of enclosure of racks or enclosures shall be "dripproof" on top. Perforated or expanded metal may be used for the back, front and sides, over areas as large as practicable, consistent with strength requirement, to promote maximum air circulation for cooling. The top shall be so constructed that water falling from directly overhead cannot enter the enclosure.

3.4.2.12 Forced air cooling.- Amplifier system assemblies shall be designed for operation in the maximum ambient temperatures existing within their rack or enclosures when the ambient temperature surrounding the rack or enclosure is 50°C. However, to prolong the operating life of various components, susceptible to such temperatures, forced air cooling facilities shall be considered in the design and shall be provided when required by the individual equipment specification.

3.4.2.12.1 Location of heat susceptible components.- Transistors, germanium and silicon diodes, and all other components adversely affected by exposure to excessive heat shall not be installed in close proximity to heat producing components such as power transformers, output transformers and power output tubes nor shall they be installed in locations where they will be exposed to hot air rising from such heat producing components.

3.4.2.13 Top bracing.- Deck mounted amplifiers and control racks shall be designed with two side mounting holes near the top and front of the rack for securing top braces when installed aboard ship.

3.5 Electrical requirements.-

3.5.1 Transformers.- All amplifier input, interstage, output, plate voltage supply and filament transformers shall be in accordance with grade 5, class S, life expectancy X of MIL-T-27. Amplifier input,



Source: https://assist.dla.mil -- Downloaded: 202610-19T09:55Z
Check the source to verify that this is the current version before use.

interstage, plate voltage supply and filament transformers shall have their primary and secondary windings isolated from each other by grounded electrostatic shields. Plate voltage supply and filament transformers shall be designed for operation from a 115 volt, 60 cycle normal power supply.

3.5.2 Relays.- Relays having contacts which control power in excess of 25 watts, all time delay type relays, and other electromagnetically operated contact making devices shall be capable of 100,000 operations at the highest rate at which they are likely to be operated in service use, but not less than two operations per minute, without requiring repair or servicing. The foregoing requirements shall be complied with while contacts are carrying loads they would normally be required to carry in the equipment. The design of relays shall be such that when mounted in their normal position the surfaces of contact springs will be in a vertical plane. Voltage surge suppression features shall be provided for the contacts of relays carrying the primary current of power amplifier plate voltage supply transformers. All relays shall be provided with removable dust-tight covers.

3.5.3 Electron tubes, transistors and metallic rectifiers.-

3.5.3.1 Electron tubes.- All electron tubes included in the design of the equipment shall be types which have been approved by NAVSEC.

3.5.3.2 Transistors and metallic rectifiers.- Transistors shall be used in the design of all preamplifiers and amplifier-oscillators. Transistors and metallic rectifiers included in the design shall be types which have been approved by NAVSEC.

3.5.4 Printed wiring.- Printed wiring may be employed for amplifier and amplifier-oscillator assemblies. However, the specific design of printed wiring boards shall be in accordance with MIL-STD-275 and MIL-STD-429 and shall be approved by NAVSEC for their application.

3.5.5 Amplifier and amplifier-oscillator assemblies.-

3.5.5.1 Power supply.- Each amplifier or amplifier-oscillator assembly shall be provided with its own power supply for all power requirements of the assembly. The power supply shall be an integral part of each assembly except where the total weight of the assembly, including the power supply, would exceed the limit specified in 3.4.2.10.

3.5.5.2 Test facilities.- A test meter together with associated multiplier and shunt resistors, and switching facilities, shall be included in each amplifier and amplifier-oscillator assembly. The foregoing test facilities shall provide a ready means for indicating the operating conditions of each amplifier, and the output signal level. The meter calibration and circuit component values shall be such that the operating condition of the amplifier or amplifier-oscillator can be determined directly from the meter without reference to tables or charts.

3.5.5.3 "Standby" and "ready" conditions.-

3.5.5.3.1 "Standby" condition.- "Standby" condition for an amplifier or amplifier-oscillator shall be defined as that condition of operation, prior to transfer to "ready" condition during which the power consumed by the equipment is reduced so as to decrease the amount of heat to be dissipated and to increase the operating life of susceptible component parts.

3.5.5.3.2 "Ready" condition.- "Ready" condition for an amplifier or amplifier-oscillator shall be defined as that condition of operation during which the equipment is fully energized and instantly available to perform its specified functions.

3.5.5.3.3 Time interval between "standby" and "ready" conditions.- Starting from a "standby" condition, the amplifier or amplifier-oscillator shall reach the ready condition within 0.5 second of the time the amplifier start circuit or an amplifier-oscillator control circuit is energized.

3.5.5.3.4 Tube life during "standby" condition.- While in the "standby" condition, power shall be consumed in such a way as to insure maximum useful life of electronic tubes and other parts consistent with the time interval requirement specified in 3.5.5.3.3.

3.5.5.3.5 Voltage surge due to transfer from "standby" to "ready" condition.- Any transient increase in voltage within an amplifier system, resulting from switching from "standby" to "ready" condition, shall not cause damage to any component within the system.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:55Z
Check the source to verify that this is the current version before use.

3.5.5.4 Power source.- All units covered by this specification shall be designed for operation from a normal power source of 115 volts, 60 cycles, single phase, having tolerances as specified in 3.3.

3.5.6 Amplifier design characteristics.-

3.5.6.1 Microphone stage input impedance.- Input impedance of the microphone stage not including the external series resistors, shall be 75 ohms (plus 10 minus 0 ohms) resistive at any frequency between 200 and 5,000 cycles per second.

3.5.6.2 Output voltage.- Amplifier output voltage to loudspeaker systems, when delivering rated output power, shall be 70 volts (plus or minus 4 volts) r.m.s. For amplifiers rated at 500 watts output, or greater, an additional output transformer tap at 95 volts (plus or minus 5 volts) shall be provided.

3.5.6.3 Output power.- Rated output power for amplifier types shall be as required by the individual equipment specification.

3.5.6.4 Amplifier gain.-

3.5.6.4.1 Preamplifier "rated gain."- With a 1000 cycle per second sine wave signal of 0.0087 volt r.m.s. applied to the input terminals of the preamplifier through two 39 ohm resistors (one resistor in each side of the line between the 0.0087 volt signal source and the input terminals) the preamplifier shall produce a signal level of 10 volts across the input of the power amplifier or an equivalent resistive load. The foregoing gain setting shall be accomplished by adjustment of the preamplifier's output level control and shall be considered "rated gain."

3.5.6.4.2 Preamplifier reserve gain.- With the output level control advanced to the maximum output position the gain of the preamplifier shall increase to a value not less than 10 db greater than "rated gain."

3.5.6.4.3 Power amplifier gain.- The gain of the power amplifier shall be such that with a 1,000 cycle per second sine wave signal of 10 volts r.m.s. applied to its input it will deliver rated output power at rated voltage.

3.5.6.5 Frequency response.- With the preamplifier set at "rated gain" and with a sine wave signal of 0.0087 volt r.m.s., at frequencies of 200, 400, 1,000, 3,000, and 5,000 cycles per second, applied to its input terminals as specified in 3.5.6.4.1, the output voltage of the power amplifier, measured across a resistive load, shall fall within the limits shown on figure 1. The foregoing requirements shall also apply when the preamplifier output level control is set at maximum output and the test input signal level reduced by an amount sufficient to cause the power amplifier to deliver rated power to the resistive load. The value of the resistive load for the above tests shall be such that it will draw rated power from the power amplifier at rated output voltage.



Source: https://assist.dla.mil -- Downloaded: 202640-19T09:55Z
Check the source to verify that this is the current version before use.

3.5.6.6 Distortion.-

3.5.6.6.1 Distortion at various frequencies.- With the preamplifier set at "rated gain" and with a variable frequency sine wave voltage (of such value as to cause the power amplifier to deliver rated power at rated voltage to a resistive load) applied to the preamplifier input terminals, the maximum harmonic distortion of the power amplifier output measured across the resistive load shall not exceed five percent for input signal frequencies of 400, 1,000 and 6,000 cycles per second. With the input signal frequency set at 200 cycles per second and with its voltage adjusted so that the power amplifier delivers power to the resistive load at a level three db below rated power, harmonic distortion shall not exceed five percent.

3.5.6.6.2 Distortion at various input levels.- At rated gain and with a 1000 cycle per second sine wave signal voltage applied to the input terminals of the preamplifier specified in 3.5.6.4.1, the total harmonic distortion of the power amplifier output voltage measured across a resistive load (of such value as to draw rated output power at 70 volts) shall not exceed the limits given for the following input voltages:

| Input voltage (volt) | Maximum harmonic distortion (percent) |
|---|---|
| 0.00087 to 0.0087 | 5 |
| 0.0174 | 10 |
| 0.0348 | 15 |

3.5.6.7 Noise and microphonics.- At "rated gain" and with a 150 ohm resistor connected across the 75 ohm preamplifier input terminals, in series with two 39 ohm resistors, the power amplifier output voltage across a resistive load (of such value as to draw rated output power at 70 volts) shall not exceed the following values when measured with an unweighted full range voltmeter. The measurement shall be made under two conditions of test — once while the component is stationary and again while the component is under vibration as specified in 3.3.

Component stationary.- Maximum measured voltage shall not exceed 0.22 volt r.m.s. (50 decibels below rated output voltage of 70 volts).
Component under vibration.- Maximum measured voltage shall not exceed 1.22 volts r.m.s. (35 decibels below rated output voltage of 70 volts).

3.5.6.8 Output voltage regulation.- At "rated gain" and with a steady sine wave signal of 0.0087 volt r.m.s. applied to the preamplifier input terminals, as specified in 3.5.6.4.1, a variation of power amplifier output loading from full rated load (a resistive load which will draw rated output power at 70 volts) to no load (open circuit) shall not cause the output voltage to exceed 99 volts r.m.s. The foregoing requirements shall apply at any frequency between 200 and 5,000 cycles per second.



Source: https://assist.dla.mil -- Downloaded: 202610-19T09:55Z
Check the source to verify that this is the currentversion before use.

3.5.6.9 Isolation against longitudinal voltages.- Longitudinal voltages shall be applied to the preamplifier input terminals in the manner described below and shall bear a ratio of not less than 50 decibels to the corresponding voltages required across the normal amplifier input terminals to produce the same amplifier output voltage. A sine wave voltage at a frequency of 1000 cycles per second shall be employed for this purpose. The longitudinal voltages shall be applied and measured as follows:

(a) Between both input terminals and ground (input terminals short-circuited).
(b) Between the junction of two 75 ohm resistors, connected in series across the input terminals, and ground.
(c) Between either input terminal and ground (input terminals open-circuited).

3.5.6.10 Stability.- The amplifier shall show no evidences of instability or oscillation when operated with the preamplifier output control set at maximum output with the preamplifier input either short or open circuited and with the power amplifier output either short or open circuited.

3.5.6.11 Overload capacity.- With the preamplifier set at rated gain and with a 1000 cycle per second sine wave signal applied to its input terminals, as specified in 3.5.6.4.1, the amplifier system shall be capable of operating continuously for a period of 10 minutes with the output of the power amplifier short circuited, without being adversely effected by the overload.

3.5.6.12 Overload limiter circuit.- The preamplifier portion of amplifier systems shall include circuits to provide the following overload limiter characteristics:

3.5.6.12.1 Overload limiter characteristics.- With a sine wave signal voltage applied to the preamplifier input terminals the increase in preamplifier output voltage resulting from an increase in the input voltage shall fall within the following limits:

| Relative preamplifier input voltage (decibels) | Relative preamplifier output voltage (decibels) |
|---|---|
| 0 | 0 |
| +6 | 0 (+1, - 0) |
| +12 | 0 (+1.5, - 0) |

Reference output voltage level (0 decibels) is 10 volts measured across the input to the power amplifier or an equivalent resistive load. Reference input voltage level (0 db) is that input voltage level required to produce reference output voltage. The foregoing requirements shall apply for frequencies of 500, 1,000, 2,000, and 4,000 cycles per second.

3.5.6.12.2 Time constant.- Under the conditions specified in 3.5.6.12.1 the preamplifier output voltage shall return to the value specified in 3.5.6.12.1 within 0.001 second after the sudden increase in input signal voltage above reference level. The preamplifier output voltage shall return to reference level in not less than 0.01 second following the sudden reduction of the input signal to reference level.

3.5.6.13 Amplifier controls.- Amplifier controls shall be provided as follows:

3.5.6.13.1 Preamplifier output level control.- An output level control shall be provided for the preamplifier. The control shall be of the continuously variable type with slotted shaft and shaft-locking nut, and shall be installed in a readily accessible location within the enclosure.

3.5.6.13.2 Power amplifier input level control.- An input level control shall be provided for power amplifiers which do not contain a preamplifier as part of the same amplifier component. The control shall be of the continuously variable type having a slotted shaft and shaft-locking nut and shall be installed in a readily accessible location within the enclosure.

3.5.7 Alarm amplifier-oscillators design characteristics.

3.5.7.1 General.- Unless otherwise specified in the individual equipment specification, two amplifier-oscillator assemblies, one to serve as a standby unit for the other, shall be included in amplifier systems



Source: https://assist.dla.mil -- Downloaded: 20261019T09:55Z
Check the source to verify that this is the current version before use.

having alarm signal requirements. Alarm signals generated by amplifier-oscillators shall be as follows:

| Signal number | Signal function |
|---|---|
| 1 | Collision alarm |
| 2 | Flight crash alarm |
| 3 | Diving alarm |
| 4 | Chemical alarm |
| 5 | General alarm |
| 6 | Flight warning alarm |
| 7 | Missile launching alarm |
| 8 | Unassigned |

The alarm signals to be made available in specific amplifier-oscillator assemblies shall be as required by the individual equipment specification for the amplifier system.

3.5.7.2 General character of signals.- The simulated signals specified in 3.5.7.3 shall be designed to be raucous in character for the purpose of making them as attention arresting as possible. For that reason the use of pure sine waves as components in the make-up of signals shall be avoided.

3.5.7.3 Generated signals.- Closure of the associated control circuits shall cause the following signals to be transmitted from the amplifier-oscillator:

| Signal number | Character of signal |
|---|---|
| 1 | A "pulsed" signal of 1,000 (plus or minus 10) cycles per second with time division characteristics as follows: Each signal "pulse" group shall consist of three "pulses" each of 0.06 seconds duration. The first two "pulses" of the group shall be followed by an "off time" of 0.06 seconds and the third "pulse" shall be followed by an "off time" of 0.3 seconds. This cycle shall be repeated continuously as long as this alarm circuit is actuated. |
| 2 | A simulated siren tone signal having its fundamental frequency swept back and forth recurrently over a range of 750 plus or minus 20 to 1,750 plus or minus 100 cycles per second at the approximate rate of 3 seconds for one full sweep. |
| 3 | A simulated motor-operated horn signal having a basic frequency of 600 plus or minus 10 cycles per second modulated at 120 cycles per second. (The harmonic content of the 600 cycle basic frequency shall be emphasized.) |
| 4 | A steady tone signal of 1,000 plus or minus 10 cycles per second. |
| 5 | A simulated single stroke gong tone signal, striking at the rate of 100 strokes per minute. |
| 6 | A "jump" tone signal alternating between 600 plus or minus 10 and 1,000 plus or minus 10 cycles per second at a rate of 1-1/2 times per second. |
| 7 | A "jump" tone signal alternating between 500 plus or minus 10 and 1,500 plus or minus 15 cycles per second at a rate of 1-1/2 times per second. |
| 8 | A "jump" tone signal alternating between 600 plus or minus 10 and 1,500 plus or minus 15 cycles per second at a rate of 6 times per second. |

3.5.7.4 Adjustment of frequency.- "Screwdriver type" adjustments with shaft-locking nuts shall be provided in each amplifier-oscillator assembly to facilitate setting the 600, 1,000 and 1,500 cycle per second signal frequencies to meet the tolerances specified in 3.5.7.3. All of the alarm signals generated by an amplifier-oscillator assembly shall be available at terminals on a terminal board located in a readily accessible location within the amplifier enclosure.



Source: https://assist.dla.mil -- Downloaded: 202610-19T09:55Z
Check the source to verify that this is the current version before use.

3.5.7.5 Duration of signals.- All signals with the exception of signal number 5 shall be transmitted continuously for as long a period as the corresponding control circuit is closed. Momentary closure of the control circuit for signal number 5 shall cause the signal to be generated for a period of 10 seconds or 15 seconds. A switch for the selection of the 10 or 15 second time intervals shall be readily accessible in the amplifier-oscillator or cabinet in which the amplifier-oscillator is installed.

3.5.7.6 Output level.- The amplifier-oscillator shall be capable of maintaining an output level of not less than 10 volts into a 600 ohm resistive load for amplifier input. This voltage shall also be used as a test signal source as specified in 3.6.6.5.1.

3.5.7.7 Adjustment of output level.- A screwdriver type adjustment with shaft-locking nut shall be provided in the amplifier-oscillator or in combination with the associated control circuits to permit independent adjustment of the amplifier-oscillator output level for the alarm signal into the amplifier input.

3.5.7.8 Noise and microphonics.- The character of the alarm signals shall not be adversely affected by the presence of any audible hum or other extraneous noises when those signals are amplified to full output level by an associated power amplifier. This requirement shall be met both with the amplifier-oscillator stationary, and under vibration in accordance with the requirements of 3.3.

3.6 Amplifier and control racks.-

3.6.1 Introduction.- This section covers the electrical facilities which are required for proper functioning of an amplifier system made up of specified numbers and types of amplifiers and amplifier-oscillator assemblies.

3.6.2 Amplifier system accommodations.-

3.6.2.1 Input facilities.- The number and type of all microphone stations or boxes associated with each amplifier system shall be as specified in the individual equipment specification.

3.6.2.2 Output facilities.- The quantity and individual designations of loudspeaker groups associated with each amplifier system shall be as specified in the individual equipment specification.

3.6.3 Amplifier system output voltages.-

3.6.3.1 System output voltage for volume indicator circuits.- The voltage available at the terminals of volume indicator circuits to microphone stations or boxes, corresponding to rated system output voltage, shall be 2.30 volts (plus or minus 0.08 volts) r.m.s. across 600 ohms.

3.6.4 Operating controls.- There shall be mounted on the front of the control panel or control rack/s, operating controls as follows:

3.6.4.1 Circuit transfer switches.- All switches employed for circuit transfer, such as amplifier transfer switches and amplifier-oscillator selector switches shall be of the JR Type as covered by Drawing 9000-S6202-73075 or 9000-S6202-73295. Wherever practicable it will be satisfactory to combine two or more switching functions on a single switch subassembly.

3.6.4.1.1 Switch markings.- All switches and operating controls shall be clearly marked to indicate the switch or control function, and to indicate all active positions or throws.

3.6.4.2 Switch and relay circuits.- Disconnect, transfer and relay switching of all power, input and output circuits shall be accomplished by opening both sides or all phases of the circuits. Single line switching will be acceptable for control circuits only.

3.6.4.3 Power indicator light.- A power indicator (pilot) light, covered by a white globe shall be provided to indicate that electric power is available at the amplifier.

3.6.4.4 Power switches.- Separate power switches shall be provided as follows:

Each amplifier channel
Each signal generator
Relay circuits



Source: https://assist.dla.mil -- Downloaded: 202610-19T09:55Z
Check the source to verify that this is the current version before use.

3.6.4.5 Amplifier input switches.-

3.6.4.5.1 Active switches.- When the amplifier design is such that more than one microphone station or box can be connected to a common amplifier channel, an amplifier input switch shall be provided for each such station or box, and shall open all circuits to that station or box.

3.6.4.6 Amplifier transfer switches.-

3.6.4.6.1 General.- When the amplifier design is such that more than one amplifier channel is employed in a system, amplifier transfer switches shall be provided to disconnect both input and output circuits of a faulty channel (or one being set up for test) and to transfer the combined system functions to the remaining channel or channels. Separate switches, having corresponding positions, may be used for input and output circuits. Those amplifier channels not selected for use shall be disconnected from all operating circuits and shall be available for test purposes, in accordance with 3.6.6.

3.6.4.7 Amplifier-oscillator selector switches.-

3.6.4.7.1 General.- Amplifier-oscillator selector switches shall be provided to permit the selection of either assembly for use with the system. The amplifier-oscillator not selected for use shall be disconnected from all operating circuits and shall be available for test purposes, in accordance with 3.6.6.

3.6.4.8 Amplifier output switches.-

3.6.4.8.1 General.- When the amplifier design is such that more than one loudspeaker group can be operated from the output of a common amplifier channel, amplifier output switches shall be provided to permit isolation of each loudspeaker group for test or repair purposes.

3.6.4.8.2 Spare switches.- The quantity of spare amplifier output switches for future system expansion will be specified in the individual equipment specification. These switches shall be connected to terminal boards and shall be wired into the amplifier and control circuits.

3.6.5 Circuit protective features.-

3.6.5.1 Amplifier main power circuits.- Both sides of the power circuits feeding each power switch specified in 3.6.4.4 shall be independently fused.

3.6.5.2 Relay power circuits.- Power to relay and control circuits, which extend beyond the amplifier enclosure (such as circuits to microphone stations or external contact makers) shall be fed through an isolation transformer to minimize the possibility of grounding the ship's power supply as a result of an accidental ground on such circuits. This requirement shall not apply to the external visual alarm circuits specified in 3.6.7.4.

3.6.5.3 Microphone equipment power circuits.- Where both sides of the power circuit are connected to a microphone station or box, each side of the circuit shall be independently fused.

3.6.5.4 Microphone audio circuits.- Individual 39 ohm protective resistors shall be connected in series with each side of each microphone audio circuit. These resistors shall be located external to the amplifier assembly, preferably in the control rack. This arrangement provides isolation as well as proper loading for each microphone.

3.6.5.5 Volume indicator circuits.- Volume indicator circuits to each microphone station or box shall be protected, so that short-circuiting the terminals of any volume indicator circuit shall not affect the amplifier output voltage, and shall not change the voltage at other volume indicator circuit terminals by more than 1 decibel.

3.6.5.6 Busy indicator circuits.- Busy indicator circuits to microphone stations or boxes shall be protected by series resistors so that short-circuiting the terminals of any busy indicator circuit shall not affect the relay power-circuit or any of the other busy indicator circuits.



Source: https://assist.dla.mil -- Downloaded: 202610-19T09:55Z
Check the source to verify that this is the current version before use.

3.6.6 Test facilities.-

3.6.6.1 General.- Test facilities shall be provided in the control rack of amplifier systems for the purpose of checking the operation of any amplifier or amplifier-oscillator assembly as an aid to system servicing and maintenance. Test circuits shall be independent of the active audio and control circuits.

3.6.6.2 Test interlocks.- Circuitry or mechanical interlocks or both shall be provided in the design of amplifiers and controls to eliminate the possibility of sounding alarm signals over loudspeakers when using test facilities to check amplifiers and amplifier-oscillators.

3.6.6.3 Amplifier start switches.- An amplifier start switch shall be provided for each amplifier channel, and when operated shall place that channel in the "ready" condition, as defined in paragraph 3.5.5.3.2. These switches shall not energize the busy indicator circuits, nor interfere with normal operation from any microphone station or box.

3.6.6.4 Amplifier-oscillator start switches.- Nonlocking type start switches shall be provided and shall make it possible to start any amplifier-oscillator, under test, transmitting any of its specified signals.

3.6.6.5 Test signal level meters and test selector switches.-

3.6.6.5.1 Input metering and switching.- An input level indicating meter and its associated input selector switch or switches shall be provided, and shall make it possible to measure the output of either amplifier-oscillator selected for test, and to connect the output of the selected amplifier-oscillator to the input of any amplifier channel selected for the test.

3.6.6.5.2 Output metering and switching.- An output level indicating meter, and its associated output selector switch or switches shall be provided, and shall permit measuring the output of any amplifier assembly selected for test.

3.6.6.5.3 Combined metering.- When required by the individual equipment specification, the metering facilities specified in 3.6.6.5.1 and 3.6.6.5.2 may be combined and a common meter used for amplifier input, amplifier output and amplifier-oscillator output level indication.

3.6.6.6 Test load resistor and switch.- A nonlocking type switch, normally in the "on" position, and associated test load resistor (of such value as to draw rated output power at 70 volts) shall be provided to permit disconnecting rated load from the output of the amplifier channel being tested.

3.6.6.7 Test input level control.- A test input level control calibrated in decibels shall be provided for controlling the test signal level of the amplifier-oscillator output (or amplifier channel input) to assist in establishing reference test input levels and to aid in making adjustments. The setting of this control shall be indicated on a fixed scale, suitably marked. Operation of this control shall not affect the system input circuits. The control shall have a range from 12 decibels above reference input level to 28 decibels below reference input level.

3.6.6.8 Relay test facility.- A relay test facility, for plug-in type relays, consisting of a test socket, indicating lamps and test switch shall be provided. The test facility shall permit testing a relay for operating coil continuity, and for contact operation in both the normal and operated conditions.

3.6.7 Control facilities.- Control circuits shall be provided to accomplish the following system functions:

3.6.7.1 Microphone press-to-talk switches.- The operation of a microphone press-to-talk switch at a microphone station or box shall:

(a) Connect the microphone audio circuit at that microphone station or box to the input of the selected amplifier channel.
(b) Apply control voltage to the loudspeaker group selector switches, if required, at that station or box.
(c) Energize appropriate busy indicator circuits.
(d) Actuate appropriate system priority circuits.
(e) Place the connected amplifier channel in the "ready" condition, as defined in 3.5.5.3.2.
(f) Connect the output of the selected amplifier channel to the selected loudspeaker groups.



Source: https://assist.dla.mil -- Downloaded: 202640-19T09:55Z
Check the source to verify that this is the currentversion before use.

(g) Disconnect the adjacent loudspeakers on the local loudspeaker circuit associated with that microphone station or box, as specified in 3.6.10.
(h) Cause the output of the Ship's Entertainment System to be reduced to a low level.

3.6.7.2 Loudspeaker group selector switches.- With the microphone press-to-talk switch operated at a microphone station or box, operation of a loudspeaker group selector switch at that station shall connect the associated loudspeaker group to the output of the selected amplifier channel.

3.6.7.3 Alarm contact makers.-

3.6.7.3.1 General.- Operation of an external alarm contact maker (consisting of a single-pole, single throw switch furnished by the installing activity) shall:

(a) Energize appropriate busy indicator circuits.
(b) Actuate appropriate system priority circuits.
(c) Place the selected amplifier-oscillator in the "ready" condition, as defined in 3.5.5.3.2, causing the required alarm signal to be transmitted.
(d) Connect the output of the selected amplifier-oscillator to the interstage input of the selected amplifier channel(s).
(e) Place the selected amplifier channel(s) in the "ready" condition, as defined in 3.5.5.3.2.
(f) Connect the output of the selected amplifier channel(s) to all connected loudspeakers, (those loudspeaker groups not actually disconnected by the loudspeaker disconnect switches, specified in 3.6.4.8).
(g) Apply full rated output voltage to all loudspeaker groups.

3.6.7.4 Additional functions for specific systems.- For specific amplifier systems, as required by the individual equipment specification, the following additional functions shall be accomplished by the operation of an alarm contact maker:

3.6.7.4.1 Visual alarm.- Operation of a Signal number 1 or number 4 contact maker shall operate a relay for control of an external visual alarm signal. The relay shall be capable of controlling a resistive load current of 7.0 amperes at 115 volts, 60 cycles, single phase. Operation of a Signal number 5 contact maker shall cause the relay to operate intermittently at the approximate rate of 1/3 second on and 1/3 second off for a period of 10 or 15 seconds.

3.6.7.4.2 Circuit SE muting.- Operation of a Signal number 1, number 3, number 4, or number 5 contact maker shall operate a relay which, through an external circuit, shall mute the output of the Ship's Entertainment System.

3.6.7.5 Loudspeaker group selection.- In general, it shall be possible for each microphone station or box (provided with loudspeaker group selector switches) on a particular announcing circuit to select any of the loudspeaker groups associated with that circuit. When required by the individual equipment specification it shall also be possible for microphone stations on one circuit to select loudspeaker groups on other circuits.

3.6.8 System priorities.-

3.6.8.1 Alarm signal priorities.- Alarm signals shall have priority over speech from microphone stations or boxes. The order of priority of the various alarm signals shall be as required by the individual equipment specification.

3.6.8.2 Speech priorities.- Unless otherwise specified in the contract or order, there shall be no priority of speech between microphone stations or boxes on the same announcing circuit. However, when two or more announcing circuits are connected to a common amplifier channel, or when two or more microphone stations, each on a different channel, are able to select the same loudspeaker group, speech priority for the microphone stations involved will be required. Speech priorities shall be as specified in the individual equipment specification.

3.6.9 Busy indicator circuits.- Busy indicator circuits shall be provided for all amplifier systems, where it is possible to connect more than one microphone station or box to a single amplifier channel. Where more than one announcing circuit is involved, each microphone station or box shall be provided with



Source: https://assist.dla.mil -- Downloaded: 202640-19T09:55Z
Check the source to verify that this is the currentversion before use.

one busy indicator circuit for each announcing circuit. These busy indicator circuits shall be energized in accordance with the following general principles:

(a) A microphone station or box shall receive a busy indication when the amplifier channel to which it is connected is in use, or when a loudspeaker group which can be selected by that station or box is in use.
(b) When more than one announcing circuit is connected to a single amplifier channel, the busy indicator circuits corresponding to these announcing circuits shall be energized simultaneously.
(c) When an alarm signal is being transmitted, all busy indicator circuits involved shall be energized.

3.6.10 Local loudspeaker cut-out circuit.- A cut-out circuit shall be included for each microphone station or box to automatically disconnect local loudspeakers, on the same system, when the microphone station or box is in use. The cut-out circuit shall consist of a normally closed relay which shall operate to open the audio circuit supplying the local loudspeakers when the microphone press-to-talk switch is operated. Power for operation of the relay shall be taken from the 115 volt, 60 cycle, single phase control power supply available in the microphone station or box. The relay shall be mounted in an enclosure and installed in the vicinity of the microphone station or box.

3.6.11 Spare control relays.- Additional relays shall be provided in the amplifier system on the basis of one relay of each type for each 25 (or fraction thereof) of that type employed, provided a minimum of five of that type are employed. These spare relays shall be for the purpose of facilitating future system revisions or modifications, shall be mounted adjacent to the active relays of corresponding type, and shall be left unconnected.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure suppliers and services conform to prescribed requirements.

4.2 Preproduction inspection.- One equipment of each type covered in the applicable specification sheet, or one equipment of a similar type, of equal or greater complexity, shall be subjected to the following preproduction inspection as applicable. Tests shall be conducted in the order listed.

| Test | Requirement Paragraph | Test Paragraph |
|---|---|---|
| Examination | ---- | 4.5 |
| Operating tests | ---- | 4.6.1 |
| Supply line voltage and frequency | 3.3, 3.5.5.4 | MIL-I-983 |
| Microphone input impedance | 3.5.6.1 | 4.6.2 |
| Output voltage and power | 3.5.6.2, 3.5.6.3 | 4.6.3 |
| Preamplifier "rated gain" | 3.5.6.4.1 | 4.6.4 |
| Preamplifier reserve gain | 3.5.6.4.2 | 4.6.5 |
| Power amplifier gain | 3.5.6.4.3 | 4.6.6 |
| Frequency response | 3.5.6.5 | 4.6.7 |
| Distortion | 3.5.6.6 | 4.6.8 |
| Noise and microphonics | 3.5.6.7 | 4.6.9 |
| Output voltage regulation | 3.5.6.8 | 4.6.10 |
| Isolation against longitudinal voltages | 3.5.6.9 | 4.6.11 |
| Stability | 3.5.6.10 | 4.6.12 |
| Overload capacity | 3.5.6.11 | 4.6.13 |
| Limiter characteristics | 3.5.6.12.1 | 4.6.14 |
| Limiter time constant | 3.5.6.12.2 | 4.6.15 |
| Generated signals | 3.5.7.3 | 4.6.16 |
| Duration of generated signals | 3.5.7.5 | 4.6.17 |
| Amplifier-oscillator output level | 3.5.7.6 | 4.6.18 |
| Amplifier-oscillator noise and microphonics | 3.5.7.8 | 4.6.19 |
| Time interval between "standby" and "ready" | 3.5.5.3.3 | 4.6.20 |



Source: https://assist.dla.mil -- Downloaded: 202640-19T09:55Z
Check the source to verify that this is the current version before use.

| Test | Requirement Paragraph | Test Paragraph |
|---|---|---|
| Voltage surge due to transfer from "standby" to "ready" condition | 3.5.5.3.5 | 4.6.21 |
| V.I. circuit voltage | 3.6.3.1 | 4.6.22 |
| V.I. circuit protection | 3.6.5.5 | 4.6.23 |
| Busy indicator circuit protection | 3.6.5.6 | 4.6.24 |
| Visual alarm circuits and circuit SE muting | 3.6.7.4.1, 3.6.7.4.2 | 4.6.25 |
| Radio frequency noise interference | 3.3 | MIL-I-983 |
| Temperature and humidity | 3.3 | MIL-I-983 |
| Enclosure | 3.3 | MIL-I-983 |
| Shock, vibration and inclination | 3.3 | MIL-I-983 |

4.3 Sampling for quality conformance inspection.-

4.3.1 Inspection lot.- All equipment of the same type presented for quality conformance inspection at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period at the choice of the inspector. Prior to presenting the lot for inspection, the contractor shall have subjected each equipment to the group A and B examination and tests and shall include in the lot presented for inspection only those equipments that have passed these tests.

4.3.2 Sampling for group A and group B examination and tests.- All equipment covered on the contract or order shall be subject to the group A and group B tests listed in 4.4 with acceptance based on 4.3.4.

4.3.3 Sampling for group C tests.- Group C tests will be required by the command or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected and subjected to each of the group C tests listed in 4.4 with acceptance based on 4.3.4.

4.3.4 Acceptance - group A, B and C tests.- The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any test, the contractor shall correct the cause of failure on future production units and shall repair the deficiency in all equipments produced on the contract or order.

4.4 Quality conformance inspection.- A sample number of equipments shall be selected and each of the selected equipments shall be subjected to the following tests, as applicable, except for group C tests which shall be conducted at a laboratory designated by the command or agency concerned.

| Group A tests | Requirement Paragraph | Test Paragraph |
|---|---|---|
| Examination | MIL-I-983 | 4.5 |
| Operating tests | ------ | 4.6.1 |
| Output voltage and power | 3.5.6.2, 3.5.6.3 | 4.6.3 |
| Preamplifier rated gain | 3.5.6.4.1 | 4.6.4 |
| Preamplifier reserve gain | 3.5.6.4.2 | 4.6.5 |
| Power amplifier gain | 3.5.6.4.3 | 4.6.6 |
| Frequency response | 3.5.6.5 | 4.6.7 |
| Output voltage regulation | 3.5.6.8 | 4.6.10 |
| Stability | 3.5.6.10 | 4.6.12 |
| Generated signals | 3.5.7.3 | 4.6.16 |
| Amplifier-oscillator output level | 3.5.7.6 | 4.6.18 |
| V.I. circuit voltage | 3.6.3.1 | 4.6.22 |

| Group B tests | Requirement Paragraph | Test Paragraph |
|---|---|---|
| Supply line voltage and frequency | 3.3, 3.5.5.4 | MIL-I-983 |
| Noise and microphonics | 3.5.6.7 | 4.6.9 |
| Limiter characteristics | 3.5.6.12.1 | 4.6.14 |
| Limiter time constant | 3.5.6.12.2 | 4.6.15 |



Source: https://assist.dla.mil -- Downloaded: 202640-19T09:55Z
Check the source to verify that this is the current version before use.

| Group B tests | Requirement Paragraph | Test Paragraph |
|---|---|---|
| Duration of generated signals | 3.5.7.5 | 4.6.17 |
| Amplifier-oscillator noise and microphonics | 3.5.7.8 | 4.6.19 |
| Time interval between "stand by" and "ready" | 3.5.5.3.3 | 4.6.20 |
| Distortion | 3.5.6.6 | 4.6.8 |

| Group C tests | Requirement Paragraph | Test Paragraph |
|---|---|---|
| Radio frequency noise interference | 3.3 | MIL-I-983 |
| Temperature and humidity | 3.3 | MIL-I-983 |
| Enclosure | 3.3 | MIL-I-983 |
| Shock, vibration and inclination | 3.3 | MIL-I-983 |

4.5 General examination.- The general examination shall be in accordance with MIL-I-983. Classification of defects shall be as specified in table I.

Table I - Classification of defects

| Categories | Defects |
|---|---|
| Critical: | |
| 1 | Assembly or portion thereof not insulated from live circuits for protection against electrical shock. |
| Major: | |
| 101 | Materials do not meet minimum requirements. Evidence such as splits, cracks, etc. indicate unauthorized material. |
| 102 | Parts not coated to prevent corrosion. Bare uncoated areas, roughness, flaking and peeling. |
| 103 | Accessibility of parts not as specified. Unable to reach parts from front of enclosure for servicing. |
| 104 | Unable to mount enclosures on bulkhead without removing parts. |
| 105 | Covers or doors not easily opened or removed; fasteners damaged, threads crossed, not captive type. |
| 106 | Heavy parts located so that center of gravity is close to top of enclosure. |
| 107 | Mounting dowels or studs not properly aligned; not sufficient to support weight of assembly. |
| 108 | Switches, operating controls, test controls, and meters not visible and accessible without opening doors or removing covers. |
| 109 | Terminal boards not in a convenient location. |
| 110 | Cable entrance not adequate, no gaskets on top cable entrance. |
| 111 | Weight of any single component in excess of 125 pounds. |
| 112 | Enclosure not "dripproof" on top. |
| 113 | Forced air cooling not provided. |
| 114 | Heat susceptible components installed too close to heat producing components such as power transformers, output transformers, and power output tubes. |
| 115 | No provision for top bracing for deck mounted equipment. |
| Minor: | |
| 201 | Preservation, packaging, and packing not as required. |
| 202 | Marking, drawing number, type, stock number and manufacturers monogram missing, incorrect, illegible or not stapled and located as specified. |

4.6 Tests.-

4.6.1 Operating tests.- In addition to applicable tests covered in 4.6.2 through 4.6.25, the equipment shall be subjected to an operating test to demonstrate satisfactory performance of components, such as switches, relays, meters and indicator lamps, relative to their normal functions in the circuit. Amplifier systems shall be subjected to a listening test to demonstrate their ability to satisfactorily handle



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:55Z
Check the source to verify that this is the current version before use.

speech signals at input levels ranging from 0.001 to 0.030 volts r.m.s. The input shall be a speech signal from a microphone or from a prerecorded magnetic tape. In either case the impedance of the test signal source shall not be greater than 150 ohms. During the test the amplifier output shall be connected to a load consisting of a loudspeaker and shunt resistor having a combined resistance value such as to draw rated power at rated voltage from the amplifier under test. The test features included in the amplifier assemblies, amplifier-oscillator assemblies and in system control facilities shall be tested to demonstrate satisfactory operation. Alarm signals shall be tested to demonstrate priority of operation as required by the individual equipment specification.

4.6.2 Input impedance.- Input impedance shall be measured to determine compliance with 3.5.6.1.

4.6.3 Output voltage and power.- Output voltage and power shall be measured to determine compliance with 3.5.6.2 and 3.5.6.3. Conditions for these tests shall be in accordance with 3.5.6.4.1.

4.6.4 Preamplifier "rated gain".- The "rated gain" of the preamplifier shall be measured to determine its compliance with 3.5.6.4.1.

4.6.5 Preamplifier reserve gain.- Preamplifier reserve gain shall be measured to determine its compliance with 3.5.6.4.2.

4.6.6 Power amplifier gain.- The gain of the power amplifier shall be measured to determine its compliance with 3.5.6.4.3.

4.6.7 Frequency response.- Frequency response shall be measured to determine compliance with 3.5.6.5.

4.6.8 Distortion.- Distortion shall be measured to determine compliance with 3.5.6.6.

4.6.9 Noise and microphonics.- Noise and microphonics shall be measured to determine compliance with 3.5.6.7.

4.6.10 Output voltage regulation.- Output voltage regulation shall be measured to determine compliance with 3.5.6.8.

4.6.11 Isolation against longitudinal voltages.- Isolation against longitudinal voltages shall be measured to determine compliance with 3.5.6.9.

4.6.12 Stability.- Stability shall be tested to determine compliance with 3.5.6.10.

4.6.13. Overload capacity.- Overload capacity shall be tested to determine compliance with 3.5.6.11.

4.6.14 Overload limiter characteristics.- Overload limiter characteristics shall be measured to determine compliance with 3.5.6.12.1.

4.6.15 Limiter time constant.- The time constant of the overload limiter feature shall be measured to determine compliance with 3.5.6.12.2.

4.6.16 Generated signals.- Generated signals shall be analyzed and measured to determine compliance with 3.5.7.2 and 3.5.7.3.

4.6.17 Duration of generated signals.- Duration of generated signals shall be measured to determine compliance with 3.5.7.5.

4.6.18 Amplifier-oscillator output level.- The amplifier-oscillator output level shall be measured to determine compliance with 3.5.7.6.

4.6.19 Amplifier-oscillator noise and microphonics.- Amplifier-oscillator noise and microphonics shall be measured to determine compliance with 3.5.7.8.

4.6.20 Time interval between "standby" and "ready".- The time interval between standby and ready shall be measured to determine compliance with 3.5.5.3.3.



Source: https://assist.dla.mil -- Downloaded: 202610-19T09:55Z
Check the source to verify that this is the current version before use.

4.6.21 Voltage surge due to transfer from "standby" to "ready" condition.- The following test shall be applied to amplifier systems to determine compliance with 3.5.5.3.5. With the power supply to amplifier system set at rated voltage and frequency and with the amplifier in "standby" condition, the press-to-talk switch (or equivalent manually or mechanically operated contactor) shall be operated, switching the system to the "ready" condition, 1,000 times at intervals of 1/2 second "on" and 1/2 second "off" followed by 1,000 operations at intervals of ten seconds "on" and two seconds "off".

4.6.22 V.I. circuit voltage.- Volume indicator circuit voltage shall be measured to determine compliance with 3.6.3.1.

4.6.23 V.I. circuit protection.- The protective feature provided for volume indicator circuits shall be tested to determine compliance with 3.6.5.5.

4.6.24 Busy indicator circuit protection.- The protective feature provided for busy indicator circuits shall be tested to determine compliance with 3.6.5.6.

4.6.25 Visual alarm circuits and circuit SE muting.- These circuit functions shall be tested to determine compliance with 3.6.7.4.

## 5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Basic equipment or item.-

5.1.1.1 Preservation and packaging.- Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.- Repair parts shall be packaged Level A, packed level C and marked level A and C respectively in accordance with MIL-E-17555.

5.2 The equipment, repair parts and tools shall be prepared for the level of shipment as specified (see 6.1).

(5.2.1 Preservation and packaging.- The equipment, repair parts and tools shall be preserved and packaged level A, packed level A and marked level A and C respectively in accordance with MIL-E-17555.)

## 6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of items required.
(c) Quantity of amplifiers or controls or both comprising an equipment.
(d) Input facilities - quantity, type, location, and associated announcing circuit of all microphone stations and boxes, including spare circuits (see 3.6.2.1).
(e) Output facilities - total quantity, individual designations, and associated announcing circuits of loudspeaker output groups, including spare circuits (see 3.6.2.2 and 3.6.4.8.2).
(f) Supply demand control center for repair parts (see 3.3.1).



Source: https://assist.dla.mil -- Downloaded: 202610-19T09:55Z
Check the source to verify that this is the currentversion before use.

(g) Quantity of repair parts (see MIL-E-17362).
(h) Quantity of technical manuals (see MIL-M-15071).
(i) Preservation, packaging, packing or marking requirements other than those required in 5.1 (see 5.2).
(j) Type of storage for equipment repair parts.
(k) Whether or not technical manuals are to be packed with the equipment.

6.2 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 This specification is used in lieu of MIL-A-20222 covering amplifiers and amplifier-control groups.

6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 5830-N007Sh)



Source: https://assist.dla.mil -- Downloaded: 20261 0-19T09:55Z
Check the source to verify that this is the current version before use.

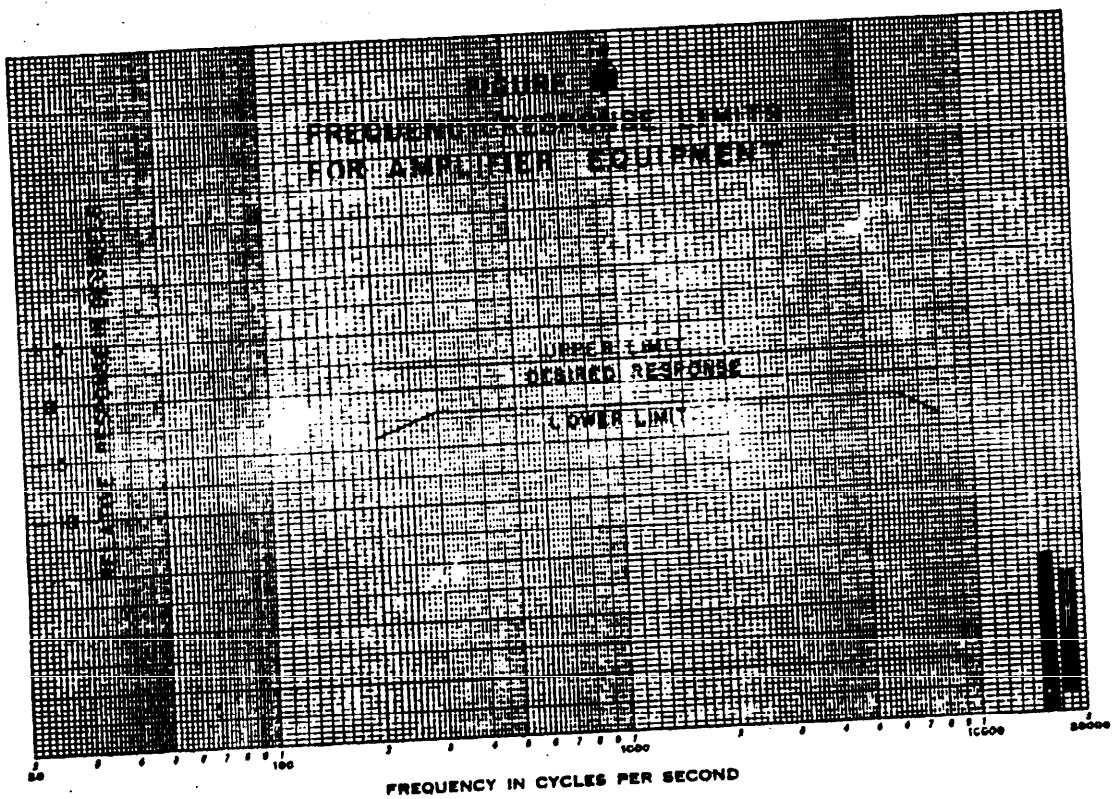



FIGURE [illegible]. [illegible] limits for frequency [illegible] for amplifier equipment.



Source: https://assist.dla.mil -- Downloaded: 20261O-19T09:55Z
Check the source to verify that this is the currentversion before use.

# INDEX


| | Paragraph | Page |
|---|---|---|
| REQUIREMENTS | 3. | 1 |
| Preproduction | 3.1 | 1 |
| Description | 3.2 | 1 |
| Equipment complement | 3.2.1 | 1 |
| General features | 3.3 | 2 |
| Repair parts | 3.3.1 | 3 |
| Mechanical requirements | 3.4 | 3 |
| Amplifier assemblies and amplifier-oscillator assemblies | 3.4.1 | 3 |
| Definition | 3.4.1.1 | 3 |
| Construction | 3.4.1.2 | 3 |
| Accessibility | 3.4.1.2.1 | 3 |
| Amplifier and control racks | 3.4.2 | 3 |
| Definition | 3.4.2.1 | 3 |
| General construction | 3.4.2.2 | 3 |
| Front servicing | 3.4.2.3 | 4 |
| Removable covers and doors | 3.4.2.4 | 4 |
| Location of heavy parts in assemblies | 3.4.2.5 | 4 |
| Mounting of amplifier and amplifier-oscillator assemblies | 3.4.2.6 | 4 |
| Mounting controls | 3.4.2.7 | 4 |
| Terminal boards | 3.4.2.8 | 4 |
| Provision for cable entrance | 3.4.2.9 | 4 |
| Weight limit | 3.4.2.10 | 4 |
| Degree of enclosure | 3.4.2.11 | 4 |
| Forced air cooling | 3.4.2.12 | 4 |
| Location of heat susceptible components | 3.4.2.12.1 | 4 |
| Top bracing | 3.4.2.13 | 4 |
| Electrical requirements | 3.5 | 4 |
| Transformers | 3.5.1 | 4 |
| Relays | 3.5.2 | 5 |
| Electron tubes, transistors and metallic rectifiers | 3.5.3 | 5 |
| Electron tubes | 3.5.3.1 | 5 |
| Transistors and metallic rectifiers | 3.5.3.2 | 5 |
| Printed wiring | 3.5.4 | 5 |
| Amplifier and amplifier-oscillator assemblies | 3.5.5 | 5 |
| Power supply | 3.5.5.1 | 5 |
| Test facilities | 3.5.5.2 | 5 |
| "Standby" and "ready" conditions | 3.5.5.3 | 5 |
| "Standby" condition | 3.5.5.3.1 | 5 |
| "Ready" condition | 3.5.5.3.2 | 5 |
| Time interval between "standby" and "ready" conditions | 3.5.5.3.3 | 5 |
| Tube life during "standby" condition | 3.5.5.3.4 | 5 |
| Voltage surge due to transfer from "standby" to "ready" condition | 3.5.5.3.5 | 6 |
| Power source | 3.5.5.4 | 6 |
| Amplifier design characteristics | 3.5.6 | 6 |
| Microphone stage input impedance | 3.5.6.1 | 6 |
| Output voltage | 3.5.6.2 | 6 |
| Output power | 3.5.6.3 | 6 |
| Amplifier gain | 3.5.6.4 | 6 |
| Preamplifier "rated gain" | 3.5.6.4.1 | 6 |
| Preamplifier reserve gain | 3.5.6.4.2 | 6 |
| Power amplifier gain | 3.5.6.4.3 | 6 |
| Frequency response | 3.5.6.5 | 6 |
| Distortion | 3.5.6.6 | 6 |




Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:55Z
Check the source to verify that this is the current version before use.

INDEX (Cont'd.)


| | Paragraph | Page |
|---|---|---|
| Distortion at various frequencies | 3.5.6.6.1 | 6 |
| Distortion at various input levels | 3.5.6.6.2 | 6 |
| Noise and microphonics | 3.5.6.7 | 7 |
| Output voltage regulation | 3.5.6.8 | 7 |
| Isolation against longitudinal voltages | 3.5.6.9 | 8 |
| Stability | 3.5.6.10 | 8 |
| Overload capacity | 3.5.6.11 | 8 |
| Overload limiter circuit | 3.5.6.12 | 8 |
| Overload limiter characteristics | 3.5.6.12.1 | 8 |
| Time constant | 3.5.6.12.2 | 8 |
| Amplifier controls | 3.5.6.13 | 8 |
| Preamplifier output control | 3.5.6.13.1 | 8 |
| Power amplifier input level control | 3.5.6.13.2 | 8 |
| Alarm amplifier-oscillator design characteristics | 3.5.7 | 8 |
| General | 3.5.7.1 | 8 |
| General character of signals | 3.5.7.2 | 9 |
| General signals | 3.5.7.3 | 9 |
| Adjustment of frequency | 3.5.7.4 | 9 |
| Duration of signals | 3.5.7.5 | 10 |
| Output level | 3.5.7.6 | 10 |
| Adjustment of output level | 3.5.7.7 | 10 |
| Noise and microphonics | 3.5.7.8 | 10 |
| Amplifier and control racks | 3.6 | 10 |
| Introduction | 3.6.1 | 10 |
| Amplifier system accommodations | 3.6.2 | 10 |
| Input facilities | 3.6.2.1 | 10 |
| Output facilities | 3.6.2.2 | 10 |
| Amplifier system output voltages | 3.6.3 | 10 |
| System output voltage for volume indicator circuits | 3.6.3.1 | 10 |
| Operating controls | 3.6.4 | 10 |
| Circuit transfer switches | 3.6.4.1 | 10 |
| Switch markings | 3.6.4.1.1 | 10 |
| Switch and relay circuits | 3.6.4.2 | 10 |
| Power indicator lights | 3.6.4.3 | 10 |
| Power switches | 3.6.4.4 | 10 |
| Amplifier input switches | 3.6.4.5 | 11 |
| Active switches | 3.6.4.5.1 | 11 |
| Amplifier transfer switches | 3.6.4.6 | 11 |
| General | 3.6.4.6.1 | 11 |
| Amplifier-oscillator selector switches | 3.6.4.7 | 11 |
| General | 3.6.4.7.1 | 11 |
| Amplifier output switches | 3.6.4.8 | 11 |
| General | 3.6.4.8.1 | 11 |
| Spare switches | 3.6.4.8.2 | 11 |
| Circuit protective features | 3.6.5 | 11 |
| Amplifier main power circuits | 3.6.5.1 | 11 |
| Relay power circuits | 3.6.5.2 | 11 |
| Microphone equipment power circuits | 3.6.5.3 | 11 |
| Microphone audio circuits | 3.6.5.4 | 11 |
| Volume indicator circuits | 3.6.5.5 | 11 |
| Busy indicator circuits | 3.6.5.6 | 11 |
| Test facilities | 3.6.6 | 12 |
| General | 3.6.6.1 | 12 |
| Test interlocks | 3.6.6.2 | 12 |
| Amplifier start switches | 3.6.6.3 | 12 |
| Amplifier-oscillator start switches | 3.6.6.4 | 12 |
| Test signal level meters and test selector switches | 3.6.6.5 | 12 |
| Input metering and switching | 3.6.6.5.1 | 12 |




Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:55Z
Check the source to verify that this is the current version before use.

INDEX (Cont'd.)


| | Paragraph | Page |
|---|---|---|
| Output metering and switching | 3.6.6.5.2 | 12 |
| Combined metering | 3.6.6.5.3 | 12 |
| Test load resistor and switch | 3.6.6.6 | 12 |
| Test input level control | 3.6.6.7 | 12 |
| Relay test facility | 3.6.6.8 | 12 |
| Control facilities | 3.6.7 | 12 |
| Microphone press-to-talk switches | 3.6.7.1 | 12 |
| Loudspeaker group selector switches | 3.6.7.2 | 13 |
| Alarm contact makers | 3.6.7.3 | 13 |
| General | 3.6.7.3.1 | 13 |
| Additional functions for specific systems | 3.6.7.4 | 13 |
| Visual alarms | 3.6.7.4.1 | 13 |
| Circuit SE muting | 3.6.7.4.2 | 13 |
| Loudspeaker group selection | 3.6.7.5 | 13 |
| System priorities | 3.6.8 | 13 |
| Alarm signal priorities | 3.6.8.1 | 13 |
| Speech priorities | 3.6.8.2 | 13 |
| Busy indicator circuits | 3.6.9 | 13 |
| Local loudspeaker cutout circuit | 3.6.10 | 14 |
| Spare control relays | 3.6.11 | 14 |
| QUALITY ASSURANCE PROVISIONS | 4. | 14 |
| Inspection procedures | 4.1 | 14 |
| Preproduction tests | 4.2 | 14 |
| Sampling and acceptance for inspection tests | 4.3 | 15 |
| Inspection lot | 4.3.1 | 15 |
| Sampling for group A and B tests | 4.3.2 | 15 |
| Sampling for group C tests | 4.3.3 | 15 |
| Acceptance for group A, B and C tests | 4.3.4 | 15 |
| Inspection tests | 4.4 | 15 |
| General examination | 4.5 | 16 |
| Operating tests | 4.6.1 | 16 |
| Input impedance | 4.6.2 | 17 |
| Output voltage and power | 4.6.3 | 17 |
| Preamplifier "rated gain" | 4.6.4 | 17 |
| Preamplifier reserve gain | 4.6.5 | 17 |
| Power amplifier gain | 4.6.6 | 17 |
| Frequency response | 4.6.7 | 17 |
| Distortion | 4.6.8 | 17 |
| Noise and microphonics | 4.6.9 | 17 |
| Output voltage regulation | 4.6.10 | 17 |
| Isolation against longitudinal voltages | 4.6.11 | 17 |
| Stability | 4.6.12 | 17 |
| Overload capacity | 4.6.13 | 17 |
| Overload limiter characteristics | 4.6.14 | 17 |
| Limiter time constant | 4.6.15 | 17 |
| Generated signals | 4.6.16 | 17 |
| Duration of generated signals | 4.6.17 | 17 |
| Amplifier-oscillator output level | 4.6.18 | 17 |
| Amplifier-oscillator noise and microphonics | 4.6.19 | 17 |
| Time interval between "standby" and "ready" | 4.6.20 | 17 |
| Voltage surge due to transfer from "standby" to "ready" condition | 4.6.21 | 18 |
| Volume indicator circuit voltage | 4.6.22 | 18 |
| Volume indicator circuit protection | 4.6.23 | 18 |
| Busy indicator circuit protection | 4.6.24 | 18 |
| Visual alarm circuits and Circuit SE muting | 4.6.25 | 18 |
| PREPARATION FOR DELIVERY | 5. | 18 |
| General | 5.1 | 18 |




Source: https://assist.dla.mil -- Downloaded: 20261O-19T09:55Z
Check the source to verify that this is the current version before use.

INDEX (Cont'd.)


| | Paragraph | Page |
|---|---|---|
| NOTES | 6. | 18 |
| Ordering data | 6.1 | 18 |
| Preproduction | 6.2 | 19 |
| **FIGURE** | | |
| Frequency Response Limits for Amplifier Equipment. | | 20 |




Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:55Z
Check the source to verify that this is the current version before use.

# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 JAN 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE
(NAVSHIPS OVPRT 12-66)

Source: https://assist.dla.mil -- Downloaded: 20261O-19T09:55Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL SHIP ENGINEERING CENTER
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER, NAVAL SHIP ENGINEERING CENTER
FEDERAL & MILITARY DOCUMENTS & DOD STANDARDIZATION BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:55Z
Check the source to verify that this is the current version before use.